

ECCS 4441 Ethical Hacking & Pen Testing

Module 2

Setting up Penetration Testing Lab



MODULE OBJECTIVES

By the end of this module, you should be able to:

1. Explain the purpose of a penetration testing lab
2. Describe the role each virtual machine plays in a penetration testing lab
3. Describe how to set up a virtual machine

PENETRATION TESTING LAB OVERVIEW (1 OF 4)

- A penetration testing lab is useful for an ethical hacker to have at their disposal
- Virtualized platforms are commonly used to create or host penetration testing labs
- The penetration testing lab provides a safe security testing environment

PENETRATION TESTING LAB OVERVIEW (2 OF 4)

Key Terms

- Sandbox – isolated software testing environment; allows safer testing of security tools that can potentially be dangerous
- Virtual machine (VM) – virtual computing environment; digital version of a physical computer
- Target – any computing object that is being pen tested
- ISO file –single file that represents the content of an entire optical disc

PENETRATION TESTING LAB OVERVIEW (3 OF 4)

The pen testing lab environment discussed throughout this text is designed for hands-on lab activities using the following virtual machines:

- Kali Linux
- Metasploitable2
- Microsoft Windows 7
- Microsoft Windows 10
- Microsoft Windows Server 2019
- Damn Vulnerable Web Application (DVWA)

The virtualized lab environment requires at least 8 GB of memory with 16 GB recommended.

PENETRATION TESTING LAB OVERVIEW (4 OF 4)

- Virtualization platform – an environment supporting virtual machines that act like computers with operating systems
- VMware – Virtual Environment
- Additional documentation and guidance is available through the course and online

SETTING UP THE KALI LINUX VIRTUAL MACHINE (1 OF 3)

Downloading and Installing VirtualBox

- Detailed instructions for installing and configuring the pen testing lab environment are presented in this course
- Additional documentation is available online

SETTING UP THE KALI LINUX VIRTUAL MACHINE (2 OF 3)

Downloading and Installing Vmware (already installed)

- Virtualization platform developed by Oracle Corp
- Available for multiple hosts and architectures
 - x86 & Intel64/AMD64
 - Windows
 - Linux & Unix
 - Mac OS

SETTING UP THE KALI LINUX VIRTUAL MACHINE (3 OF 3)

- Kali Linux is widely known and is the preferred platform for pen testing activities
- Free and freely available
- Available in OVA and ISO formats

DISCUSSION ACTIVITY 1

Some may argue that creating a ready-to-use hacking distribution, such as Kali Linux, is arming potential threat actors with tools to perform malicious activities against potentially vulnerable targets.

Others may argue that a freely available pen testing toolkit like Kali and offering it for use allows security engineers and those in blue team roles to more efficiently gain access to the tools malicious attackers have at their disposal.

Discuss your arguments for and against the open access to pen testing tools to the tech community at large.

SETTING UP TARGETS (1 OF 7)

- After Kali Linux is configured and running in VMware, six target virtual machines will be created
- Installation and configuration steps are similar for target VMs
- Care should be taken to ensure pen testing lab virtual machines are properly created to maintain isolation from the host computing environment

SETTING UP TARGETS (2 OF 7)

Metasploitable2

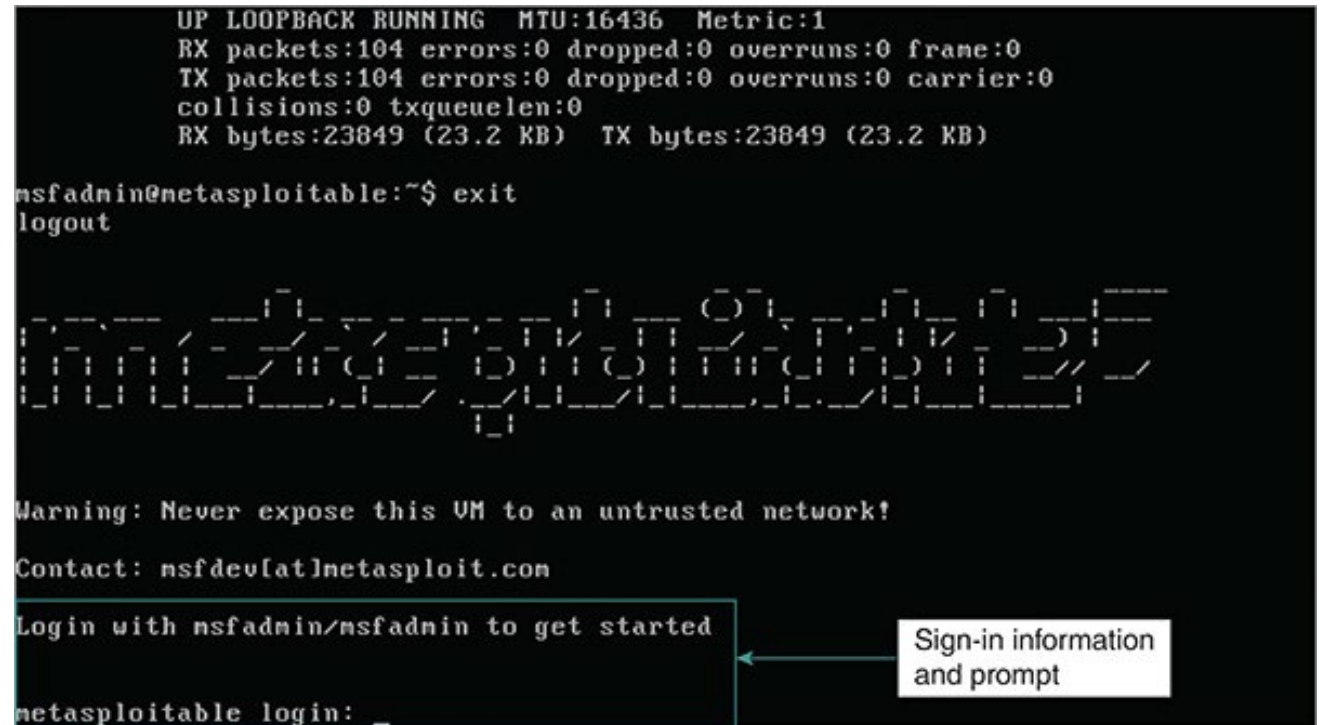
- Provided and owned by Rapid7
- Purposely constructed vulnerable security testing platform
- Virtual machine files supplied individually, not as an appliance platform
- Built on Ubuntu Linux

```
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:104 errors:0 dropped:0 overruns:0 frame:0
TX packets:104 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:23849 (23.2 KB) TX bytes:23849 (23.2 KB)

msfadmin@metasploitable:~$ exit
logout

Warning: Never expose this VM to an untrusted network!
Contact: nsfdev[at]metasploit.com

Login with nsfadmin/nsfadmin to get started
metasploitable login: _
```

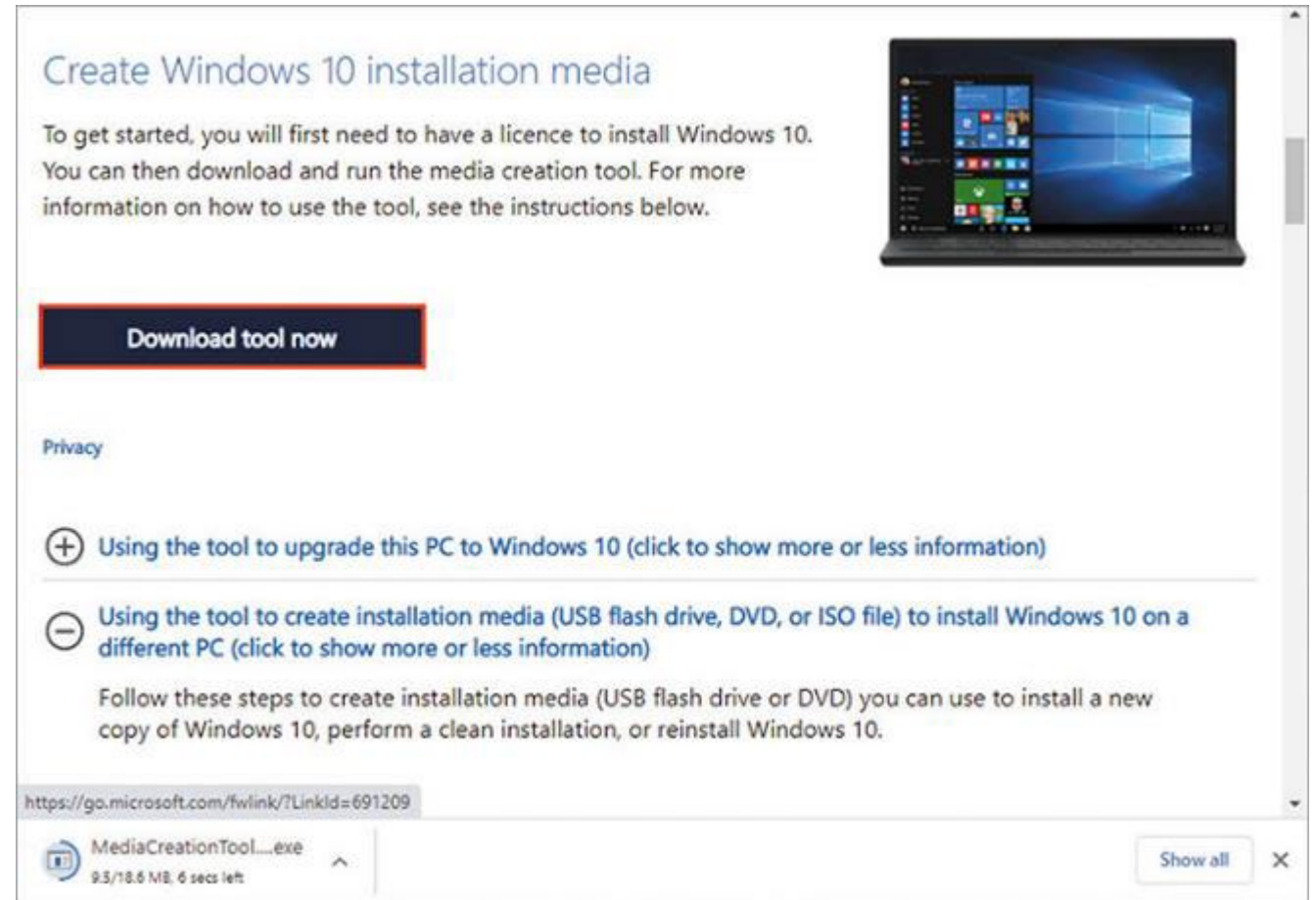


Metasploitable login screen with login info

SETTING UP TARGETS (3 OF 7)

Microsoft Windows 7

- End of product life January 2020
- 12% market share as of book writing
- Inherently vulnerable due to lack of support

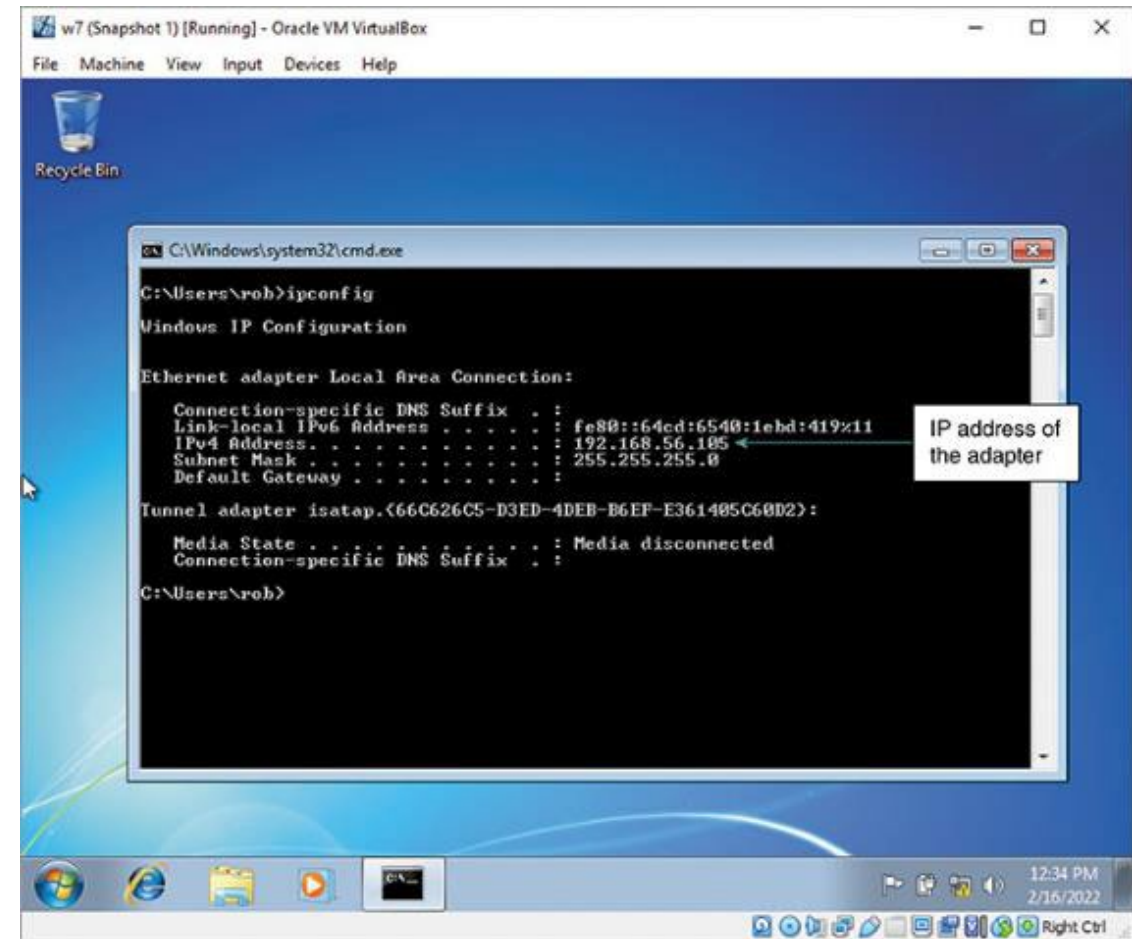


Downloading the Media Creation Tool for Windows 10

SETTING UP TARGETS (4 OF 7)

Microsoft Windows 10

- Runs on 1.3 billion devices worldwide
- Commonly used and encountered during pen tests

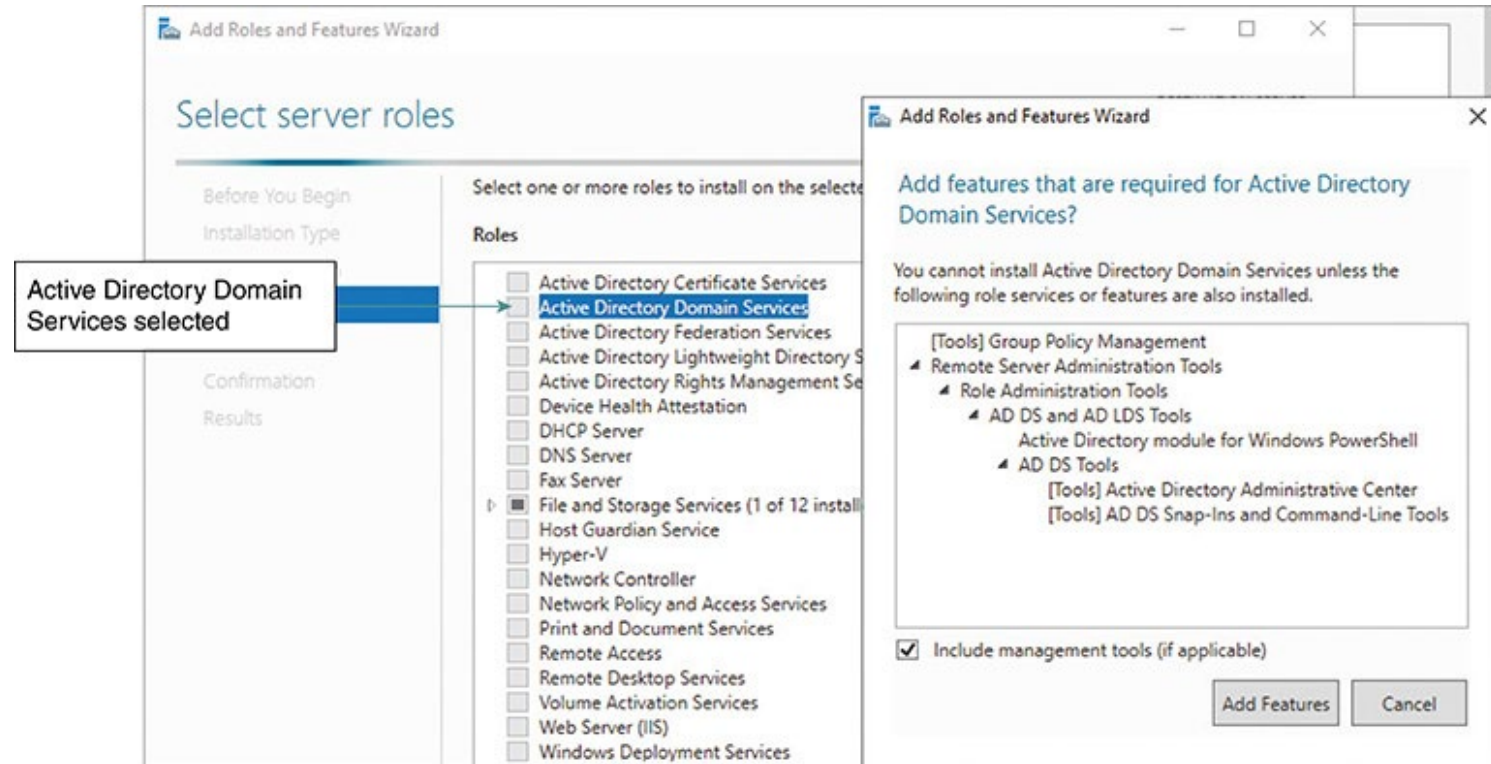


Windows 10 as virtual machine

SETTING UP TARGETS (5 of 7)

Windows Server 2019

- Functions as Domain Controller running Active Directory
- Microsoft owns 48% market share as of book writing
- Trial license available for 180 days

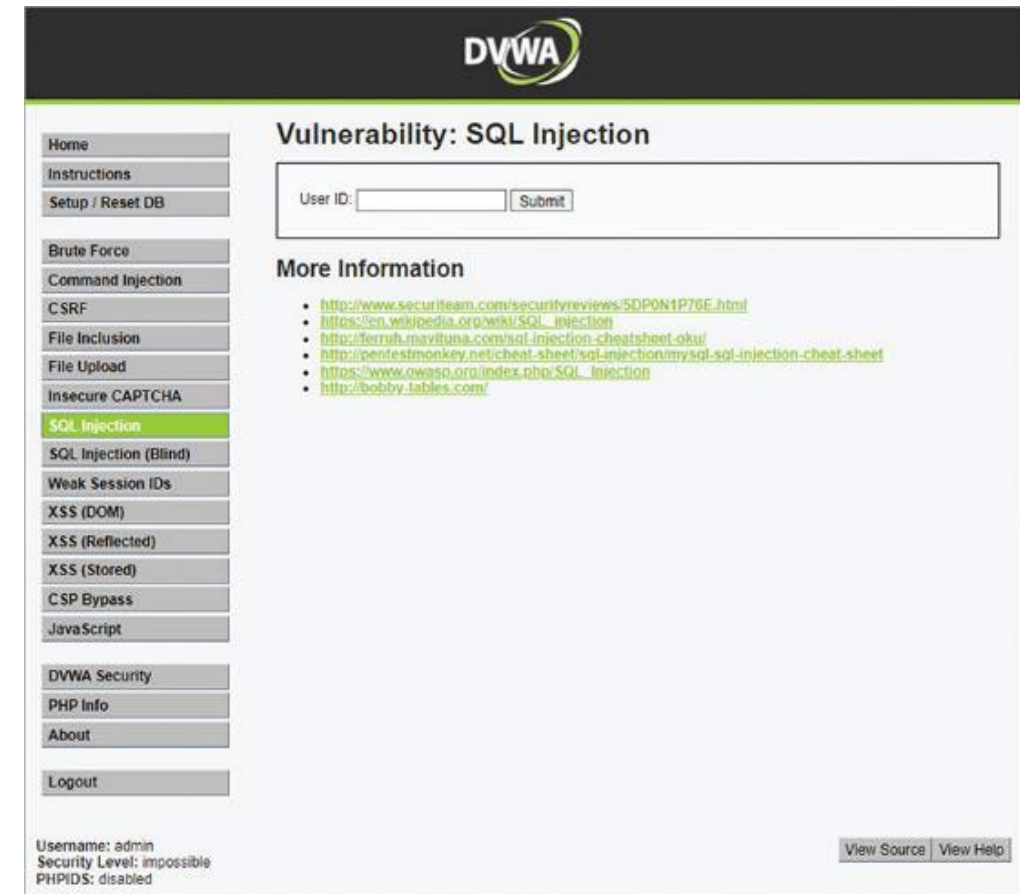


Add Roles and Features Wizard in Server 2019

SETTING UP TARGETS (6 OF 7)

Damn Vulnerable Web Application (DVWA)

- Intentionally vulnerable web app platform for security testing
- Runs on Linux operating system
- Freely available



DVWA open in a web browser

SETTING UP TARGETS (7 OF 7)

Metasploitable 3

- Purposely constructed a vulnerable security testing platform
- Built on Ubuntu Linux and Windows Server. We will use Windows Server based Metasploitable 3

SELF ASSESSMENT

A penetration testing lab is very useful tool for aspiring ethical hackers and professional pen testers alike to have access to. Such a lab lists the steps needed to build a penetration testing lab like the virtualized lab built in this module.

What components are required to build a safe, reliable, and isolated penetration-testing lab for conducting ethical hacking practice exercises like those provided in this course? What steps are necessary in building the pen test lab?

SUMMARY

Now that the lesson has ended, you should be able to:

- Explain the purpose of a penetration testing lab
- Describe the role each virtual machine plays in a penetration testing lab
- Describe how to set up a virtual machine